

The Measurement of Muscle Hypertrophy

chapter

3

A variety of measurement techniques are available for assessing muscle hypertrophy. It is important to note that these methods merely provide estimates of muscle mass, and all have limitations that must be considered when attempting to draw conclusions from their use. The gold standard reference in the determination of muscle morphology is cadaver analysis; thus, *in vivo* techniques can never meet the highest degree of accuracy (95). That said, reasonable predictions can be gleaned from other assessment methods, with some demonstrating better accuracy than others.

This chapter focuses on describing how to employ various assessment techniques to determine muscle size and its change over the course of regimented exercise programs. Thus, the discussions of the underlying concepts on which these techniques are based are brief; if you seek greater insights into this topic, refer to the authoritative text, *Human Body Composition* (53). In this chapter, methods are categorized based on whether the assessment of muscle is made indirectly or in a site-specific fashion. Indirect methods include skinfold measurement, hydrodensitometry, air displacement plethysmography (ADP), dual X-ray absorptiometry (DXA), and bioelectrical impedance analysis (BIA); site-specific measures include circumference measurement, ultrasound, computerized tomography (CT), magnetic resonance imaging (MRI), and muscle biopsy.

Indirect Measures

Indirect measures of body composition analysis can be categorized into one of three basic models: two component (2C), three component (3C), or four component (4C). As the name implies, 2C models partition the body into two distinct compartments: fat mass and fat-free mass (30, 54). *Fat-free mass* includes all nonfat components, including skeletal muscle, body organs, bone, and body fluids. The 2C model is predicated on assumptions that the density of fat-free mass and the ratio of protein to mineral remains constant—an assumption that may not always hold true. Examples of 2C models include skinfold measurement, hydrodensitometry, and ADP.

The major issue when attempting to extrapolate findings from 2C models to determine changes in muscle mass is that other nonfat components, particularly water, may influence fat-free mass readings. The water content in the human body can vary from day to day and also over longer periods. This is especially relevant to women during the phases of their menstrual cycle, where cyclical hormonal alterations can cause total body water to fluctuate greatly. Moreover, resistance training-associated changes in fat-free mass hydration are routinely observed, with results at least partially attributed to alterations in blood volume (89). Finally and importantly, considerable

interindividual variability exists in the density of fat-free mass, particularly in the ratio of water to mineral, which can confound results in an athletic population (82).

The 3C model uses the 2C approach of partitioning the body into fat mass and fat-free mass, and then it further segments the fat-free mass portion into protein and mineral fractions. Although the 3C model is more sophisticated than the 2C model, it is still limited by an inability to distinguish body water from other fat-free tissue proteins. DXA is an example of a 3C model.

Four-component models are considered the gold standard in indirect body composition analysis. This model segments the body into fat, protein, mineral, and water. Moreover, it eschews assumptions of constancy in the hydration component of fat-free mass, as well as the ratio between protein and mineral content. This is particularly important when assessing body composition changes over the course of a weight-loss protocol because water loss from adipose tissue is otherwise reported as a loss of fat-free mass (3). Multiple measurement methods are needed to devise a 4C model; the preference is to use reference standards that target each of the components. Common 4C components include hydrodensitometry, DXA, and BIA.

Although multicompartiment models are considered true criterion methods for body composition assessment (93), it is important to note that their findings reflect overall fat-free mass, and observed changes are thus not specific to individual skeletal muscles or muscle groups. This poses several limitations. For one, the magnitude of hypertrophy can differ between muscles over the course of an exercise program. Different protocols from 4C models might show similar increases in hypertrophy, but differences may exist in how certain muscles responded compared to others. Thus, if one protocol found greater hypertrophic increases in the upper-body musculature while another showed greater increases in the lower-body musculature, the results may not be reflected in 4C findings. Moreover, the 4C model would have limited value in programs that target a specific muscle or muscle group. For example, it is

common in research to employ protocols that target only the upper arms or the upper legs and often just one muscle in a region (e.g., triceps or biceps; quadriceps or hamstrings). Therefore, as with all indirect measures, 4C models are generally best employed in combination with site-specific methods to provide a better understanding of musculoskeletal dimensions and their changes over time.

What follows is an overview of these indirect hypertrophy assessments: skinfold measurement, hydrodensitometry, DXA, ADP, and BIA. The discussion will include the basic procedures, advantages, disadvantages, and associated research supporting their validity for assessing muscle mass and hypertrophic changes that may occur through exercise regimens.

Skinfold Measurement

The skinfold technique is perhaps the most basic indirect method for obtaining estimates of fat-free mass. The very low cost and convenience of skinfold measurement makes it a popular choice as a field assessment, and its predictions are more accurate than those based on circumference measurement (9).

Skinfold measurements are obtained on the right side of the body in sequential fashion. Common skinfold sites include the chest, midaxilla, triceps, subscapular, abdomen, suprailium, thigh, biceps, and calf (figure 3.1). Valid results can be obtained using three-, four-, seven-, and nine-site assessments. At least two measurements are taken at each site, and results are averaged to enhance accuracy. Once final values are acquired, the sum of the skinfolds is entered into one of many regression equations to generate an estimate of body composition.

Accuracy of the skinfold technique is highly dependent on the skill of the practitioner. Its biggest potential downside is a lack of testing competency, leading to technical measurement error. Proper training and many hours of practice on a wide variety of individuals are necessary for becoming proficient in the technique. The quality of the calipers is another possible source of error in skinfold analysis. Calipers can differ in dial accuracy, surface area of the jaw faces, and force or pressure exerted at the



FIGURE 3.1 A person taking a skinfold assessment.

jaw faces (76), and these differences can have a profound impact on results (16). As a general rule, inexpensive calipers provide less accurate measurements than more costly units. Another potential issue with skinfold measurement involves interindividual differences in skin qualities and subcutaneous fat patterning, which can result in an inability to palpate the fat-muscle interface and thus unduly bias results (9). Finally, prediction formulas are specific to a particular sample of the population and on factors such as age, sex, nutritional status, genetic background, and activity levels; the assumptions used to create these equations, which are based on the premise that fat-free mass is similar in a limited sample of a given population, may not be universally valid and thus result in errors of estimation (9).

Despite its limitations, skinfold measurement generally provides valid assessments of

fat-free mass at a single time point. Its predictive capacity in bodybuilders was found to be highly correlated with the 4C model ($r = .93$) (89). Lean mass estimates by skinfold also showed high agreement with hydrodensitometry ($r = .94$) in a cohort of NCAA Division I football players (70).

When assessing fat-free mass changes in male athletes over a 7-day creatine monohydrate loading protocol, seven-site skinfold measures showed a strong correlation with hydrodensitometry ($r = .93$) (48). Compared to a 4C model, however, results have been mixed. Skinfold measurement was well correlated with 4C ($r = .88$) for estimating fat-free mass changes in bodybuilders over an 8-week resistance training program; these values were higher than those obtained from a 3C model (89). Alternatively, both the three- and seven-site skinfold methods underestimated changes in fat-free mass (by -2.5 to 2.7 kg, or -5.5 to 5.8 lb) compared with a 4C method in elite male judo athletes at baseline and before competition (82), indicating potentially meaningful differences.

Equations have been developed to estimate muscle cross-sectional area from the combination of circumference and skinfold measurements. Although these methods show reliability compared to a CT scan, they significantly underestimate cross-sectional area changes during regimented resistance training (18). That said, the errors in results are relatively consistent and therefore track changes in a pattern that is similar to CT (18), making the assessment a viable tool for evaluating muscle mass over time.

KEY POINT

The skinfold technique can be a practical method for assessing muscle development. However, the accuracy of results will be highly dependent on the practitioner's proficiency, and, with this in mind, findings must be interpreted skeptically. Combining skinfold measurements with circumference measurements is a viable means for tracking changes in muscle cross-sectional area, and the same caveats apply based on the skill of the practitioner.

Hydrodensitometry

Hydrodensitometry (also known as underwater weighing) is a well-established method for estimating lean tissue mass (figure 3.2). It is often considered the gold standard of 2C models and for this reason is frequently used as a criterion reference for validating other measures and is commonly included in 4C models. The concept is based on Archimedes principle, which states that a body's weight under water is directly proportional to the volume of water displaced by the body's volume. Given that lean tissue is denser than water, and fat mass is less dense, a person with more body fat necessarily will have greater buoyancy and thus weigh less under water than someone with a greater amount of lean tissue.



FIGURE 3.2 An underwater weighing tank.

A limitation of hydrodensitometry is the need for a dedicated water tank, making it less accessible than other methods of assessment. Data are obtained by completely submerging a person in the water tank while he or she is attached to a scale and then calculating body volume as the difference between dry weight and underwater weight, with corrections made for water density. Body density is then derived by dividing body mass by body volume. Various formulas can be employed to estimate lean mass from the data. Although the Siri and Brozek equations have been most commonly used for prediction, these formulas are based on findings from white male and female cadavers and thus may not accurately estimate results for other races and ethnicities (45).

Hydrodensitometry has several potential sources of error that can negatively influence determination of lean mass. For one, the person being tested must maximally exhale and remain completely motionless throughout testing; deviations from this protocol can substantially alter results. In addition, corrections must be made for residual lung volume; relatively modest differences (600 ml) in this variable can affect body composition measures by as much as 8%, and the magnitude of error rising with increasing body size (45).

Hydrodensitometry is generally accurate in estimating lean mass across populations. It shows an almost perfect correlation ($r = .99$) with the 4C model when comparing lean mass in bodybuilders at a single point in time;

KEY POINT

Hydrodensitometry can be considered a useful tool for estimating lean mass. However, as with every 2C model, its results reflect all fat-free components. Accordingly, if used to assess hypertrophic changes associated with regimented resistance training, the values obtained would not necessarily represent changes in muscle mass because alterations in other components (e.g., water) may obscure the ability to draw valid inferences.

PRACTICAL APPLICATIONS**BIOCHEMICAL ASSESSMENT OF MUSCLE MASS: A VIABLE METHOD OF HYPERTROPHY MEASUREMENT?**

Several biochemical methods can be employed to estimate total body skeletal muscle mass. Although not as well-known as other methods of measurement, biochemical techniques can be viable options for morphological assessment. The most studied of these methods involves the measurement of 24-hour urinary creatinine excretion, which shows a good correlation with muscle mass in humans (44). The approach originated from animal research showing congruity between total body creatine and urinary excretion of creatinine. However, the validity of the method depends on the assumption that conversion of creatine to creatinine is constant among and within individuals. While this may have credence in a sedentary population, it does not necessarily hold true during resistance training, in which values can vary widely. Moreover, validity of the technique is predicated on accurate urine collection over the entire study period as well as stable maintenance of dietary protein intake, which can be challenging under free-living situations.

A novel technique using deuterated creatine, referred to as the D_3 -creatine dilution method, has been proposed as a superior alternative to other biochemical approaches. D_3 -creatine is a stable isotope that is consumed orally in water and taken up almost exclusively by skeletal muscle, which represents the storage site of approximately 95% of the body's creatine stores (muscle has no capacity for creatine synthesis). The total-body creatine pool is then calculated from D_3 -creatinine enrichment in urine via liquid chromatography–tandem mass spectrometry; an estimate of muscle mass is obtained by dividing the creatine pool size using an estimation of creatine content in muscle.

The D_3 -creatine dilution method was initially validated in rodents, showing a strong relationship with estimates of lean tissue mass ($r = .96$) (83). A follow-up study demonstrated the method could be employed repeatedly over time to accurately measure lean tissue changes in growing rats (84). Collectively, the findings show promise for D_3 -creatine dilution as an assessment tool.

Subsequent pilot work in humans reported good agreement in muscle mass estimates between D_3 -creatine dilution and MRI ($r = .87$) (13). The method showed less bias than DXA-derived estimates, which overestimated muscle mass by approximately 2-fold compared to MRI. However, the study was carried out in a hospital setting with a 5-day period of continuous urine collection; hence, its applicability to free-living populations is extremely limited.

In an attempt to enhance practicality, a study was undertaken to determine whether a urine sample obtained after an overnight fast would suffice to accurately predict muscle mass, both as a single measurement and for longitudinal assessment (14). Correlation analysis showed strong agreement between the D_3 -creatine dilution method and MRI-derived predictions of muscle mass ($r = .88$ to $.91$). However, the method underestimated muscle mass by 2.3 to 3.0 kg (5.1 to 6.6 lb) as well as produced significantly greater intraindividual variability than MRI. Moreover, variances were greater in women than in men, calling into question its validity when assessing in women.

(continued)

Biochemical Assessment of Muscle Mass: A Viable Method of Hypertrophy Measurement? *(continued)*

Overall, emerging evidence indicates that D_3 -creatine dilution is a potentially useful tool in determining muscle morphology. However, measurements are specific to the entire body; it is not possible to glean insights into regional muscular dimensions. In addition, and perhaps more important from a hypertrophy standpoint, validity of the method is dependent on a stable population, which precludes its use in exercise protocols. Therefore, its applicability seems to be limited to sedentary populations and may be most relevant for assessing sarcopenia.

moreover, its accuracy remains high when assessing changes in lean mass over the course of an 8-week resistance training program ($r = .91$) (89). When compared with single-point 4C assessment across 54 validation studies, hydrodensitometry underestimated body fat percentage by a relatively small 0.1% to 1.2%, which implies only a slight overestimation of lean mass. Results tend to be less accurate for premenopausal women, whose values obtained through this technique vary widely across the phases of their cycles. These fluctuations can be partly explained by changes in total body water, but other factors also seem to play a role (12). To minimize potential error, premenopausal women should be assessed at the same time within their cycle for repeated measurements.

Air Displacement Plethysmography

ADP estimates body composition using principles similar to those of hydrodensitometry. However, as the name implies, it uses air displacement (as opposed to water displacement) along with pressure–volume relationships to estimate body volume. The concept is consistent with Boyle’s law, which states that the volume of a gas is inversely proportional to the pressure exerted by the gas at a fixed temperature.

ADP is commonly carried out in an egg-shaped, fiberglass unit called a Bod Pod (figure 3.3). The testing procedure is relatively simple:

The person sits motionless inside the Bod Pod chamber, a test is performed, and the unit derives an estimate of body composition based on a predetermined algorithm. A benefit of ADP is its convenience, taking only about 10 minutes to complete after calibration of the unit. Also, by eliminating the need to be submerged under water, the Bod Pod provides a more inviting alternative to hydrodensitometry. Because of these attributes, it is often used as a reference method in 4C models.

Several factors can alter the validity of ADP measurements. First and foremost, the measurement is affected by body hair, body temperature, moisture, and clothing (28, 71). Hence, valid results require individuals to wear a scalp cap and tight-fitting swimsuit and, if applicable, shave body and facial hair. Modifications in body position during testing also can affect results, albeit to a relatively small degree (72). As with other indirect assessment modalities, physical activity and food and fluid intake must be restricted before testing to prevent confounding the analysis. Finally, the Bod Pod unit is sensitive to alterations in room temperature; readings can be significantly altered by modest deviations in the surrounding environment (29).

In general, ADP produces results similar to those of hydrodensitometry for determining body composition at a single time point over a wide array of populations (80). However, its accuracy is somewhat compromised in



FIGURE 3.3 A Bod Pod, which carries out ADP.

comparison to 3C and 4C models. A study in collegiate male hockey players found that ADP overestimated fat-free mass compared with DXA (19). Similar overestimations have been observed when comparing Bod Pod to a 4C model (47). The extent of these differences is relatively modest, but nevertheless may be potentially meaningful when attempting to discern the practical implications of research.

ADP generally has not fared well in predicting changes in body composition across longitudinal studies. The technique showed a strong correlation to hydrodensitometry ($r = .95$) when assessing fat-free mass changes in athletes consuming creatine monohydrate over a 7-day period. On the other hand, estimates of fat-free mass using ADP showed a poor correlation with DXA ($r = .34$) in a cohort of overweight men and women in response to an 8-week weight loss program (96). Moreover, a study tracking body composition across a 6-month weight loss program in overweight men and women found no correlation at all between the Bod Pod and DXA for predicting lean mass changes (34). The combined findings indicate ADP is not a viable assessment method in this population. A dearth of longitudinal research comparing ADP

KEY POINT

ADP has merit for assessing lean mass at a given time point but may lack accuracy when attempting to estimate changes over the course of a training program. Limited longitudinal data in athletes suggest it can be a valid option, but conclusions are difficult to interpret because of the lack of comparison with reference methods in this population.

with 3C and 4C reference methods in athletic populations precludes the ability to draw strong inferences as to its efficacy.

Dual X-Ray Absorptiometry

DXA is a 3C model that divides the body into total bone, lean soft-tissue mass, and fat mass (figure 3.4). The technique has gained popularity as a method to measure body composition in research-based settings, largely because of its combination of high reliability, convenience, and relatively low per-scan cost. Although DXA emits radiation, the associated amounts are very low and generally not considered an issue with infrequent use. However, a licensed radiological technician is often required to operate the unit, making it less practical in these circumstances.

A distinct benefit of DXA is its ability to estimate segmental lean mass for the arms, trunk, and legs. Validated equations have been developed to calculate skeletal muscle mass from DXA-derived data (49), although the algorithm was found to have an unexplained systematic bias that overestimates the variable in Caucasians (38). Nevertheless, DXA is often considered a surrogate measure for hypertrophy and was recently deemed the reference standard for measuring muscle mass by a European Society for Clinical and Economic Aspects of Osteoporosis and Osteoarthritis working group on frailty and sarcopenia (11).

Numerous potential sources of error are associated with DXA. For one, the type of unit used can influence findings; the level of accuracy varies between pencil- and fan-beam



FIGURE 3.4 (a) A DXA unit and (b) a DXA image of lean body mass.

(a) Doncaster and Bassetlaw Hospitals/Science Source; (b) courtesy of Dr. Grant Tinsley.

units (45). In addition, accuracy also varies by population. Factors that can introduce bias include sex, size, fatness, and disease state. The magnitude of these biases is approximately ≤ 2 kg (4.4 lb) of fat-free mass (97), which may be practically meaningful. Moreover, although DXA is less prone to fluctuations in total body water compared to hydrodensitometry (45), it nevertheless is quite sensitive to dietary manipulation. In particular, increased glycogen stores and the associated water it attracts can increase bodyweight by several kilograms, potentially having a profound effect on results. A recent study found that food consumption, irrespective of macronutrient content, increased DXA estimates of total and regional lean soft-tissue mass up to 1.7% and 3%, respectively, with results holding true for both men and women (88). Thus, an overnight fast is needed to prevent confounding caused by nutrient intake.

When measurements are obtained at a single time point, DXA displays very high correlation with the 4C model for estimating lean mass in bodybuilders ($r = .97$) (89). DXA also performs adequately in comparison to site-specific measures in this regard. Thigh skeletal muscle area

as determined by single-slice CT showed a good correlation with DXA-derived measures of thigh fat-free mass ($r = .86$); correlations with thigh muscle volumes derived from multislice CT were even higher ($r = .98$) (51). Comparisons with MRI have produced correlation coefficient values ranging from .86 to .97 (85), indicating strong relationships between modalities.

While DXA's predictive ability is fairly high, it demonstrates a reduced ability to track changes in markers of muscle mass over the course of a regimented exercise program. When tracking changes in a cohort of bodybuilders across an 8-week resistance training program, it demonstrated a fairly high correlation with 4C ($r = .77$), although this relationship was significantly lower than when assessed at a single time point (89). A study in elite judokas reported similar findings; DXA results explained only 38% of the 4C reference values for fat-free mass when tracking the athletes from a period of weight stability to just before competition. Agreement with changes in lean mass as assessed by site-specific measures is even more compromised; the correlation coefficient was just .49 compared to MRI-derived

KEY POINT

DXA is perhaps the best indirect method for assessing lean mass. Given its convenience and relatively low risk, it is a good option for estimating muscle development at the group level. Nevertheless, it is not sensitive to alterations in body water, and lacks the ability to detect subtle changes over the course of a regimented resistance training program. Ideally, it should be used in combination with site-specific measures to provide greater insights into muscle tissue composition (51).

values of midhigh volume (85). Similarly, although both DXA and CT scans were able to detect significant increases in thigh mass over a 10-week strength training program, a potentially meaningful difference of approximately 2 kg (4.4 lb) was noted between methods. Other research lends support to these findings (20). The discrepancies between single-point measures and those taken over the course of a longitudinal training study are likely caused by confounding from variations in adipose tissue or skin mass over time, which alters how the DXA algorithm determines lean mass (51).

Bioelectrical Impedance Analysis

BIA emits a low-level electrical current throughout the body and then estimates body composition based on the ease of electrical flow through the body while taking into account the individual's height, weight, and sex. BIA is perhaps the most convenient of all body composition modalities, taking less than a few minutes to complete. It also is relatively inexpensive and thus is employed extensively in research settings. Higher-end BIA units are able to estimate segmental lean mass and skeletal muscle mass. A unique feature of certain BIA units is an ability to calculate intracellular and extracellular fluid compartments, which cannot be gleaned from other techniques.

BIA can be classified into two basic categories: single frequency and multifrequency.

Single-frequency units commonly emit a 50 kHz electrical impulse that not only passes through extracellular bodily fluids but also may traverse the cellular membrane (10, 50). The limitations in making estimations from a single 50 kHz frequency in combination with potential confounding from intracellular penetration of the electrical current compromise the validity of these units (10, 57). Alternatively, multifrequency units apply currents below 50 kHz to quantify extracellular water with minimal penetration of the cellular membrane as well as currents up to 1,000 kHz that estimate the quantity of total body water and thus allow for the calculation of intracellular water (10) (figure 3.5). Because of these advanced capabilities, multifrequency BIA units are purported to be less prone to error than single-frequency units, and thus may represent a superior choice for predicting body composition (37, 64, 86).

Regardless of frequency capacity, BIA has several potential sources of error. For one, it



FIGURE 3.5 A multifrequency BIA unit.

Photo courtesy of InBody USA.

is highly susceptible to hydration status and fluctuations in body fluids; this can be especially problematic in premenopausal women when measurements are obtained at different times of the menstrual cycle, although research generally shows minimal effects in this regard (15, 39, 60). Moreover, as with DXA, food consumption can substantially alter readings (88), necessitating an overnight fast for accuracy. In addition, the algorithms can vary in sophistication between manufacturers and units; this is important because accuracy of results depends on the predictive equation on which calculations are based. In particular, significant variations in predictions are seen when the algorithm does not properly account for inherent differences in body composition between populations (e.g., general public versus athletes) (94). Finally, electrode configurations can vary between units. Hand-to-hand, foot-to-foot, and hand-to-foot are common arrangements, and differences in analysis between these designs can influence body composition results (88).

Because of the vast differences between methods and units, it is somewhat difficult to draw general conclusions about the validity of BIA in measuring fat-free mass. Single-frequency BIA was shown to correlate well with the 4C model in single-point assessment of lean mass in bodybuilders (89). Single-point comparisons between multifrequency BIA and DXA generally show agreement between modalities, although BIA tends to overestimate fat-free mass (23, 26, 37, 92). In regard to segmental lean mass analysis, BIA tends to slightly underestimate fat-free mass compared to reference methods, although agreement is relatively good at the group level (94).

When evaluating changes in hypertrophy-related measures over time, BIA shows somewhat mixed results. A high correlation was found between BIA and hydrodensitometry ($r = .92$) when assessing changes in fat-free mass in a cohort of male athletes consuming creatine monohydrate over a 7-day loading period (48). Good agreement was demonstrated between multifrequency BIA and DXA ($r = .71$) for lean mass changes in young men across a 10-week resistance training program; however, the two methods were poor predictors of each other

KEY POINT

BIA can be a valid option for assessing lean mass, although accuracy depends on the sophistication of the given unit and its algorithms. While BIA theoretically has an enhanced ability to determine skeletal muscle mass by parceling out the effects of body water, the validity of these measures have not been well studied.

in regard to changes in segmental measurements (77). Alternatively, compared to DXA, multifrequency BIA overestimated fat-free mass in a cohort of resistance-trained men and women following a 4-week hypoenergetic diet; however, the differences were relatively small (0.18 to 0.25 kg, or 0.39 to 0.55 lb) and not statistically significant (5). Moreover, a study comprising middle-aged and older females found that while DXA was able to identify a small increase in lean tissue mass, the change went undetected by multifrequency BIA (81). The type of BIA unit also must be taken into account; a multifrequency, hand-to-foot unit showed a substantially higher correlation with DXA ($r = .66$) than a single-frequency foot-to-foot unit ($r = .30$) following a 6-month weight loss program (34).

Table 3.1 summarizes the advantages and disadvantages of indirect hypertrophy measures.

Site-Specific Measures

As the name implies, site-specific assessments evaluate muscle morphology at a given aspect of a muscle. Site-specific measures can be obtained either *in vitro* (biopsy), whereby muscle is assessed on a microscopic level, or *in vivo* (circumference measurement, ultrasound, CT, MRI), whereby muscle is assessed on a macroscopic level. *In vitro* measures allow analysis of the size of individual fibers, whereas *in vivo* measures assess gross muscle morphology. *In vivo* measures can be assessed in a single dimension (muscle thickness), which looks at a given point of a portion of a muscle; in two dimensions, which looks at a given slice of a

TABLE 3.1 Advantages and Disadvantages of Indirect Hypertrophy Measures

Modality	Advantages	Disadvantages
Skinfold	• Convenient • Inexpensive • Noninvasive • Can be combined with circumference measures to estimate CSA	• Requires highly skilled practitioner for good validity • As a standalone assessment only estimates FFM, not muscle mass • Does not measure region-specific changes
Hydrodensitometry	• Good validity • Results not influenced by operator	• Largely confined to research and laboratory settings • Burdensome • Only estimates FFM, not muscle mass • Does not measure regional-specific changes in FFM • Relies on assumed densities of FFM and FM compartments
ADP	• Generally good validity for single-point measures • Not burdensome • Results not influenced by operator	• Largely confined to research and laboratory settings • Requires restricted nutrition and exercise before testing and body hair prep to prevent confounding • Not as accurate as hydrodensitometry • Relies on assumed densities of FFM and FM compartments • Does not measure regional-specific changes in FFM • Validity compromised for estimating changes over time
DXA	• Good validity • Differentiates between bone and lean soft-tissue mass • Can estimate regional changes in FFM • Not burdensome • Relatively inexpensive • Results not influenced by operator	• Often requires operation by a licensed radiological technician • Affected by changes in glycogen status • Requires restricted nutrition and exercise before testing to prevent confounding • Lacks the ability to detect subtle changes in lean soft-tissue mass over time • Differences within and between manufacturers and units can affect validity • Small dose of radiation limits repeated use
BIA	• Generally good validity • Not burdensome • Relatively inexpensive • Can estimate regional changes in FFM • Can estimate intracellular and extracellular body water composition • Results not influenced by operator	• Accuracy depends on regression equations employed by a given manufacturer • Requires restricted nutrition and exercise before testing to prevent confounding • Differences within and between manufacturers and units can affect validity

Abbreviations: CSA = cross-sectional area; FFM = fat-free mass; FM = fat mass; ADP = air displacement plethysmography; DXA = dual X-ray absorptiometry; BIA = bioelectrical impedance analysis.

RESEARCH FINDINGS**DISCERNING BETWEEN INDIVIDUAL AND GROUP RESULTS**

As with results in all applied exercise-related studies, it is important to understand the difference in the implications of body composition research when studying a group versus an individual. Research findings generally are reported as the collective mean (i.e., average) of data obtained from a cohort of subjects. However, there will be a variance in responses across different subjects, the extent of which is exhibited as the standard deviation of the data. Sometimes these variances can be quite large. Hence, simply taking the mean at face value does not necessarily reflect the results a given person can expect to see.

In the case of body composition testing, a given method of assessment can produce a wide range of values in lean mass changes, both positive and negative, and end up showing no mean change. However, from a mean statistical standpoint, the findings can be virtually identical to another assessment tool showing a much narrower range of values that hover around the zero mark. These issues have been demonstrated in numerous validation studies for hypertrophy assessments. For example, ADP and hydrodensitometry were found to be interchangeable with the 4C model for quantifying group changes in fat-free mass when using a diet-induced weight loss program; however, large differences were seen on an individual level, leading the authors to conclude: “as indicated by the wide limits of agreement . . . individual estimates of changes in body composition when taken alone should be interpreted with caution” (55). Similarly, a study that investigated agreement between 4C, 3C, and 2C models found all three methods gave acceptable group mean values, but on an individual level, only the 3C model served as a valid alternative to 4C (89).

Group versus individual discrepancies also may apply to sources of error associated with a given assessment tool. For instance, alterations in food intake increased DXA estimates of total and regional lean soft-tissue mass up to 1.7% and 3%, respectively, on a group level; on an individual level these increases were >4.5% and 9%, respectively (88). From an individual standpoint, a person therefore potentially could expect to experience much larger variations in body composition measure from dietary alterations than that seen for the cohort as a whole.

The greatest interest in body composition assessment for most individuals generally involves tracking lean mass changes over time. In this regard, it is relatively unimportant (within reason) whether a given assessment tool may over- or underestimate lean mass compared to a reference standard at a given time point. Most relevant is that the method of choice is reliable, providing consistent measurements from one assessment to the next. For example, single time point analysis shows multifrequency BIA overestimates fat-free mass by 0.58 to 0.84 kg (1.3 to 1.8 lb) compared with DXA (61). However, if one’s primary objective is to determine the extent to which fat-free mass changes in conjunction with an exercise program, these differences essentially would be moot provided the same absolute magnitude of error persists on a consistent basis. That said, there is evidence that the magnitude of error may change with alterations in body composition (25, 55, 74), presumably because the density of fat-free mass is altered with changes in body weight, violating the assumption that it remains constant. Monitoring large changes in muscle mass using indirect measurement tools therefore should be treated more skeptically when examining individual-level data.

portion of a muscle (muscle cross-sectional area); or in three dimensions, which accounts for hypertrophic changes across the entire muscle (muscle volume). The ramifications of these classifications are important because they encapsulate different constructs, and therefore can lead to different conclusions drawn from the same data (42).

Although one-dimensional assessments can be used in hypertrophic analysis, their usefulness is limited because muscle size may increase differently in different dimensions. For example, width may increase differently than length, and this difference would not be observed in muscle thickness measurements.

Accordingly, two-dimensional measures can provide greater insights into muscle morphology compared to one-dimensional assessments. Two primary classifications of two-dimensional measurements are anatomical cross-sectional area (ACSA) and physiological cross-sectional area (PCSA). ACSA is operationally defined as the cross-sectional area of its muscle obtained perpendicular to its longitudinal axis. Alternatively, PCSA represents the cross-sectional area of a muscle obtained perpendicular to its fibers. Because the fibers of nonpennate muscles are all parallel to the long axis, ACSA and PCSA produce similar results. However, the fibers in pennate muscles such as the deltoids, rectus femoris, and triceps brachii are aligned orthogonally, and thus values between the constructs (ACSA versus PCSA) can vary widely when assessing these muscles.

Three-dimensional assessments allow for the determination of muscle volume. This can be accomplished with MRI or CT by taking multiple serial scans along a muscle's length and then integrating them as a function of distance to obtain volume (42). Alternatively, three-dimensional ultrasound affords the ability to stitch frames together with the aid of motion capture technology, then transform the frames to create a singular image of the entire muscle (42). In general, muscle volume provides the most accurate depiction of muscle size at the whole-muscle level. However, by considering muscle growth as a whole, volume estimates discount the potential importance of a given strategy to elicit regional-specific changes in

muscle mass. For example, a 10-week program of isolated eccentric and concentric resistance exercise reported similar increases in muscle volume between conditions (31). However, evaluation of hypertrophic changes along the length of the muscle found greater distal growth with eccentric training, whereas the growth in the midportion was greater with concentric training. Collectively, the findings indicate a benefit to combining the two types of action when the goal is maximal muscle development. This emphasizes the relevance of reporting both total volume and hypertrophy of specific regions of a muscle (e.g., proximal, mid, and distal aspects) to develop a complete understanding of the hypertrophic effects of a given program.

Another important point is that macroscopic measures do not provide insight into potential fiber type-specific changes in muscle growth. Some strategies appear to promote greater hypertrophy in Type I or Type II fibers (40), which may indicate a synergistic benefit of combining various training approaches. Different training strategies also may elicit differences in exercise-induced sarcoplasmic versus myofibrillar hypertrophy (41). Thus, both site-specific imaging and histology are needed to derive a complete perspective on the impact of a training program on muscle development.

The following is an overview of these site-specific hypertrophy assessments: circumference measurements, ultrasound, CT, MRI, and biopsy. As in the section on indirect hypertrophy measures, these sections discuss the basic procedures, advantages, and disadvantages associated with each method and the research on the validity of drawing morphological inferences about muscle mass.

Circumference Measurements

Circumference (girth) measurements are by far the crudest site-specific assessment for hypertrophy. The technique involves placing a flexible tape measure, generally made of cloth, over an area of the body and measuring its girth (figure 3.6). Measurements are commonly taken at the midpoint of the body area, but any site along the muscle can be assessed.



FIGURE 3.6 A person taking a circumference measurement.

Reliability of circumference assessments is acceptable and is better than that reported with skinfold testing (9). However, the technique has several serious limitations that call into question its utility for measuring muscle mass. First, unlike other site-specific methods that directly measure a given muscle, circumference assessments measure a general area of the body. So, while other methods can, for example, directly assess the size of the biceps brachii and vastus lateralis, circumference measures the entire upper arm or thigh. As such, a girth measurement of the chest could be confounded by growth of the latissimus dorsi. Second, and most importantly, the circumference method cannot distinguish between different tissue components within the area of measurement. Changes in fat mass, skin thickness, muscle tissue, and intra- and extracellular water will be reflected in the measure, thus compromising construct validity (9).

Ultrasound

Ultrasonography is a noninvasive technique that uses sound waves emitted from a transducer to assess muscle morphology (figures 3.7 and 3.8). A transducer is coated with gel

KEY POINT

Circumference measurements are poorly suited to the evaluation of muscle development. The method provides only gross estimates of hypertrophy and, as a standalone assessment, results must be viewed skeptically. However, combining the technique with other methods can help glean greater insights into human muscle mass and its changes over time.

and then placed over the muscle of interest. The transducer emits sound waves that penetrate bodily tissues; these waves are partially reflected back to the unit and displayed as an image. When a satisfactory scan is obtained, the sonographer freezes the image and obtains the desired measurement. The image can then be stored to a hard drive for future retrieval and comparison. The modality is very safe; the same technology is routinely used to track fetal development over the course of a pregnancy. Moreover, scanning can be carried out relatively quickly and inexpensively. Thus, ultrasound is the most popular site-specific modality employed in hypertrophy research.

Ultrasound has two primary imaging modes: A-mode (amplitude modulation) and B-mode (brightness modulation). A-mode involves the emission of a narrow sound beam to scan tissue discontinuity; the resultant output takes



FIGURE 3.7 An ultrasound unit.

SIGRID GOMBERT/Science Source

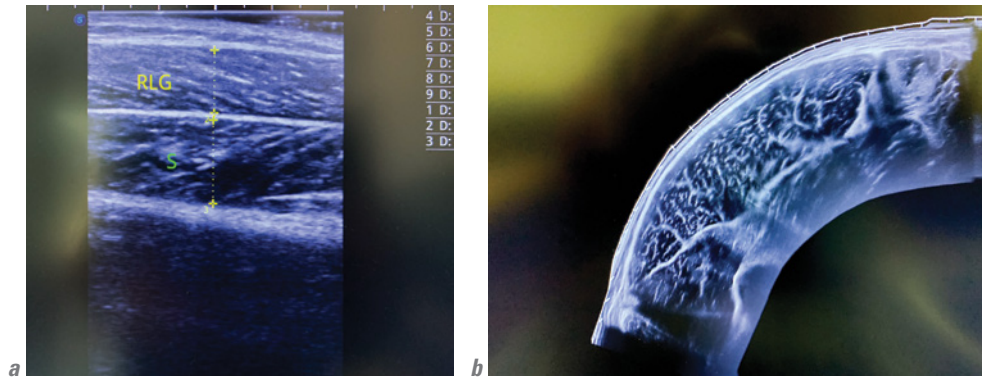


FIGURE 3.8 Two ultrasound images of muscle showing (a) muscle thickness and (b) cross-sectional area.

Brad Schoenfeld (both photos)

the form of spikes on a graph. Alternatively, B-mode emits a linear array of sound waves to generate images of the tissue of interest, and hence produces resolution superior to that of A-mode. Although A-mode has been used in research to assess muscle morphology, its validity for this purpose has not been well-studied. Thus, further discussion will be specific to B-mode imaging, which has substantial support in the literature as a valid assessment tool.

Ultrasound scans can be obtained either one dimensionally by measuring the thickness of a muscle or two dimensionally by measuring a muscle's cross-sectional area. For muscle thickness, measures are obtained at a single site along the muscle belly; multiple sites can be scanned to obtain insight into potential regional-specific adaptations that may have taken place. The fairly recent introduction of extended field-of-view ultrasound represents a technological advance in the ability to achieve high-quality cross-sectional imaging. These techniques involve the sonographer manually sweeping the transducer across the muscle at a regulated rate of speed. The software algorithms of the ultrasound unit then merge the sequential frames collected during real-time scanning to reconstruct a panoramic two-dimensional image. Three-dimensional ultrasound imaging is an emerging technology that shows promise for accurately estimating muscle volume (6); however, research into its validity for assessing hypertrophic changes over time is scant.

The biggest threat to internal validity (i.e., the trustworthiness of the data) when using ultrasonography is the skill of the operator. A high level of proficiency is required to obtain reliable, high-quality images. Differences in the amount of pressure that the sonographer exerts on the transducer against the skin can result in substantial variations in measurements, resulting in a high degree of inter-rater error. That said, measures can be highly reproducible with appropriate training (33). Another potential issue with ultrasound is that when tracking changes in muscle size over time, scans must be taken at the same site in each testing session; deviations in where measurements are obtained along the line of the muscle skew results. Finally, it is more difficult to obtain high-quality images in some individuals than in others and in some muscles than in others. The reasons for these discrepancies are not always apparent, but lean individuals tend to be easier to scan than those carrying higher levels of body fat in the region of interest. In regard to scanning differences between muscles, some research indicates the posterior thigh is more difficult to visualize than the anterior thigh, potentially because of the geometry of the femur (2).

Overall, high intra-rater reliabilities for ultrasound have been reported for the lower extremity muscles, with the majority of intraclass correlation coefficients (ICCs) ranging from .90 to .99 (2). Studies show good inter-rater

reliability as well, with ICCs between .82 and .95. Ultrasound also displays good accuracy for muscle thickness measurements obtained at a single time point. Compared to cross-sectional area as determined by MRI or CT, ultrasound measures of anterior thigh muscle thickness showed correlation coefficients of .76 to .98, with the majority of reported values $>.90$ (2). Similar findings have been shown for the lower leg, with good to strong correlations noted between ultrasound and CT or MRI findings (correlation coefficients of .70 to .91). Single-point assessments of quadriceps cross-sectional area using panoramic ultrasound also show good to excellent agreement with MRI (4, 78) and CT (69); however, its agreement was found to be poor when assessing cross-sectional area of the gastrocnemius (78).

Findings are somewhat disparate when comparing changes in muscle development over time between ultrasound and MRI. A study involving 6 weeks of blood flow restriction training concluded that ultrasound measures of muscle thickness produced similar conclusions about hypertrophy as MRI-derived measures of cross-sectional area; however, the estimates of the magnitude of change were not equivalent (52). In another study, pre- to poststudy ultrasound-derived measures of vastus lateralis muscle thickness obtained at 50% of femur length showed a moderate to strong correlation with MRI-derived measures of ACSA ($r = .69$) following 12 weeks of isokinetic resistance training for the knee extensors (32). However, correlation between ultrasound measures and muscle volume determined by MRI in the same study was poor ($r = .33$) (32). Discrepancies in findings seemingly can be explained by differences in regional hypertrophy that are routinely seen in the quadriceps across training studies (8, 66, 67). Additionally, extended field-of-view ultrasound imaging showed high agreement with MRI in detecting changes in muscle cross-sectional area (ICC = .929, SEM = .94 cm²) over the course of a 21-week resistance training program (4).

Equations have been developed to estimate muscle volume from ultrasound imaging. A formula combining measures of muscle thickness along with limb length was found

KEY POINT

Ultrasound testing presents an efficient method for assessing muscle hypertrophy and displays good accuracy when carried out by a qualified sonographer. Obtaining scans at multiple sites along the length of a given muscle can provide greater insight into overall development of that muscle. This is particularly relevant in the quadriceps musculature, which routinely shows distinct regional intramuscular adaptations in response to regimented resistance training.

to be a reasonably good predictor of muscle volume compared to MRI, with coefficients of determination varying from 41.9% for the knee extensors to 70.4% for the elbow flexors (63). However, values were obtained at a single time point, and given regional-specific differences in hypertrophy that occur with regimented exercise (8, 56, 66, 67), the accuracy of the formula for assessing hypertrophic changes in muscle volume over time seems suspect.

Computerized Tomography

CT uses X-ray-based technology to produce cross-sectional images of a given body area, including muscles (figures 3.9 and 3.10). Along with MRI, it is considered a reference method for evaluating muscle morphometry.

Scanning is carried out in a doughnut-shaped device with a table through the middle. The



FIGURE 3.9 A CT unit.

DR P. MARAZZI/Science Source

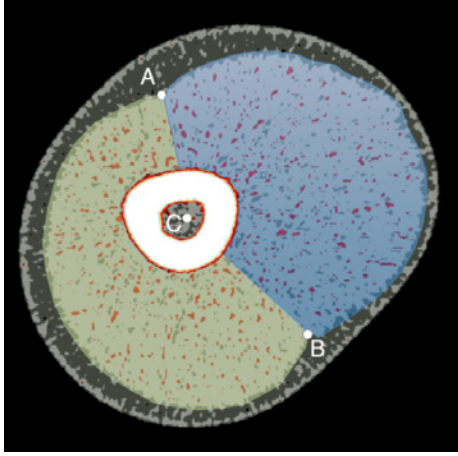


FIGURE 3.10 A CT image of a muscle.

Courtesy of Dr. Tom Maden-Wilkinson and Dr. Alex Ireland.

person lies supine on the table, and an X-ray tube rotates around the muscle of interest, emitting ionizing radiation beams. The emitted X-ray beams are attenuated while crossing the muscle and then received by the machine's detectors to produce thin-sliced images, which are processed with the aid of algorithms that define each CT slice. Two-dimensional images of the muscle's cross-sectional area are rendered, with pixels related to tissue density.

While CT provides excellent insights into muscle morphology, it is expensive (although less so than MRI) and inconvenient. Moreover, its emission of relatively high levels of ionizing radiation is not considered safe for repeated measurements (73). Thus, the practical value of CT is limited for assessing hypertrophy.

CT has consistently shown high reproducibility for the determination of human muscle size (58) and generally displays good agreement when validated against cadaver values (62). However, although CT is considered a reference method for measurement of muscle mass, its accuracy is somewhat less than MRI. Whereas MRI-derived cross-sectional area assessments were shown to be within $\pm 7.5\%$ of cadaver analysis, those of CT systematically overestimated measurements by 10% to 20% (22). Moreover, validity is further compromised in muscles that have closely apposed muscle bellies, which can increase difficulty in determining intermuscular boundaries (22).

KEY POINT

CT imaging is an excellent option for evaluating muscle hypertrophy, and it can be considered a reference standard. However, it is costly and generally only available in a hospital-based setting. Importantly, it cannot be used for frequent measurements because of the high dose of radiation emitted per scan. Thus, CT has limited practical use as a tool in assessing muscle morphology in the general population.

With respect to muscle volume, CT of the legs shows very high agreement with DXA measures of leg fat-free mass ($r = .98$) (51). However, as previously noted, the validity for DXA in determining highly accurate measurements of regional lean mass remains somewhat suspect. Studies examining the validity of CT's ability to measure changes in muscle volume are lacking.

Magnetic Resonance Imaging

MRI is widely regarded as the gold standard for assessing muscle hypertrophy because it displays better soft-tissue contrast than CT (figures 3.11 and 3.12). The technique is noninvasive and does not produce ionizing radiation (as with CT), making it a safe and relatively com-



FIGURE 3.11 An MRI unit.

STEPHANE DE SAKUTIN/AFP via Getty Images

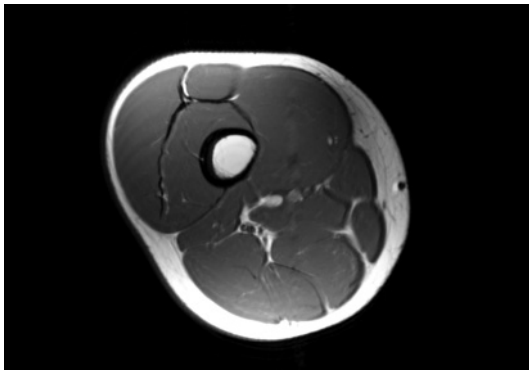


FIGURE 3.12 An MRI image of a muscle.

Courtesy of Dr. Martino Franchi.

portable measurement option. However, its high cost is a major drawback, limiting widespread use in practice. Moreover, MRI units are bulky and require controlled environments, thereby limiting accessibility.

The MRI scanning process generally involves lying supine in a tube-like machine and remaining motionless for the duration of the test. The MRI unit creates a magnetic field inside the muscle of interest, causing protons to align in the magnetic field. The protons are then activated by a pulsed radio frequency, causing them to absorb energy. The protons then release energy as the pulse is discontinued, and the protons return to their original position. The released energy is detected by the machine, facilitating image acquisition (73). Image resolution is generally excellent, providing distinct delineation of muscle borders that allows for precise measurements of muscular dimensions.

MRI has been validated *in vivo*, showing very high correlations in muscle cross-sectional area measures of the arms and legs when compared to cadavers (7, 22, 62). However, as with CT, accuracy can be somewhat compromised when subjectively interpreting boundaries between muscles in close proximity to one another, especially muscles with multiple heads (22).

MRI also demonstrates the ability to accurately determine measures of whole-muscle volume, although results are not as impressive as with cross-sectional area estimates. The technique has been validated for this purpose across a wide array of muscles for both the upper (21,

KEY POINT

MRI is the preferred choice for evaluating muscle hypertrophy and can be considered the reference method for validating other techniques. However, as with CT, its high cost and lack of accessibility make it impractical for widespread use.

87) and lower (79) body. However, the volume of certain muscles demonstrates a greater ability to be accurately assessed than others; one study reported differences between MRI and dissection measurement ranging from 7.7% to 21.6% (21). In particular, muscles with high ratios of surface area to volume may predispose them to segmentation error, compromising the validity of measurement (21). Moreover, volumes of smaller, shorter muscles tend to be underestimated, likely due to an inability to obtain a sufficient number of samples along the muscle's length. The adductor brevis, for example, showed a large measurement error between MRI and cadaver analysis, with results attributed to the fact that the muscle was visible in only 3 of the 12 images sampled (79). Thus, the ability of MRI to accurately estimate muscle volume is highly dependent on the number of serial images obtained along the longitudinal axis of a given muscle. Although no studies have validated MRI-derived measures of muscle volume over the course of a longitudinal exercise study, it seems reasonable to conclude that its precision would be similar to single-point analysis.

Muscle Biopsy

Site-specific measures of muscle size can be assessed at the microstructural level via tissue biopsy (figures 3.13 and 3.14). The process involves making an incision in the region of interest, and then inserting a needle into the incision site and extracting a small amount of muscle tissue. The extracted tissue is cut into thin slices and analyzed microscopically. Various techniques can be employed to stain the samples for assessment of fiber cross-sectional area and fiber type-specific cross-sectional area



FIGURE 3.13 A researcher taking a muscle biopsy.

Courtesy of Mark Tarnopolsky, MD, PhD.

as previously described in the literature (42). Biopsies also can be used to analyze differences between protein subfractions (e.g., sarcoplasmic versus contractile) within extracted tissue, providing further insights into the composition of exercise-induced muscle hypertrophy.

Despite its unique and wide-ranging capabilities for assessing muscle hypertrophy, the biopsy method has several drawbacks. First and foremost, it is an invasive technique. Although the biopsied area is anesthetized, some discomfort is experienced during the procedure; depending on individual pain tolerance, the discomfort can be onerous. In addition, biopsies are specific to a very small region of muscle tissue (~100 mg, or the size of a pencil eraser). As previously mentioned, hypertrophy can manifest in a nonuniform manner along the length of a muscle, and the results obtained by a single biopsy do not necessarily reflect those of the entire muscle. Moreover, differences exist in the location of fibers within a muscle, with Type I fibers tending to reside deeper into the muscle than Type II fibers (17, 65) and both fiber types generally displaying a larger size in the deeper muscular regions (43); thus, the depth of needle penetration can influence findings.

Differences generally exist between the reported magnitude of hypertrophy from biopsy measures and those obtained from site-specific methods, although there are exceptions (59).

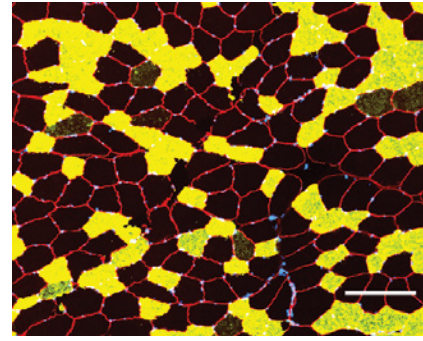


FIGURE 3.14 A microscopic image of a muscle biopsy (cross-sectional area).

Courtesy of Dr. Michael Roberts.

When compared to whole-muscle quadriceps cross-sectional area measures via MRI, most studies show higher increases in muscle fiber cross-sectional area with biopsy (24, 27, 91, 98), although some have reported lower values (67). Similar discrepancies are seen when biopsy is compared to CT, again generally biasing to a higher increase in quadriceps cross-sectional area (35, 68, 90). The relative differences in the magnitude of findings between micro- and macroscopic measures often are substantial. For example, Verdijk and colleagues (90) reported increases in Type II quadriceps cross-sectional area of 28% as obtained via biopsy, while CT measures showed just an 8.5% increase. Results from Frontera and colleagues (36) were almost identical, with 28% increases shown by biopsy and only 10% increases by CT. Correlational analysis shows a moderate association ($r = .58$) between resistance training–induced changes in biopsy-derived fiber cross-sectional area versus that obtained for the quadriceps cross-sectional

KEY POINT

Although valuable hypertrophy-related inferences can be gleaned from biopsies, the limitations of the method make it somewhat flawed as a standalone measure. Its greatest value lies when used in combination with site-specific measures because it provides unique insights into hypertrophic changes occurring at the microstructural level.

area as a whole by MRI (1). Regardless of the differences in magnitude of changes, the vast majority of studies show that microscopic and macroscopic methods track in parallel with one another. Limited research on changes in upper-body cross-sectional area seem to show congruity with lower-body findings (75).

Table 3.2 shows the advantages and disadvantages of site-specific hypertrophy measures.

Conclusion

Many methods for measuring and estimating muscle hypertrophy are available. Of the indirect measures, 2C models are limited by their insensitivity to changes in body water. However, they can serve as valuable assessment tools in the proper context and with a comprehension of their limitations. DXA is perhaps the most

TABLE 3.2 Advantages and Disadvantages of Site-Specific Hypertrophy Measures

Modality	Advantages	Disadvantages
Circumferences	• Convenient • Inexpensive • Noninvasive • Can be combined with skinfold measures to estimate CSA	• Measures a general region of the body, and thus estimates of a given muscle (e.g., triceps brachii) are confounded by growth of other muscles (e.g., elbow flexors) • Cannot differentiate changes in fat mass from FFM
Ultrasound	• Very good validity when performed by experienced personnel • Relatively inexpensive • Convenient • Noninvasive • Safe • Can be used to take multiple measures along a muscle • Can estimate muscle thickness, CSA, and/or volume	• Not as accurate as MRI or CT • Validity is highly dependent on the skill of the practitioner • Not readily able to estimate total-body muscle mass
CT	• Excellent validity • Noninvasive • Can be used to measure CSA and/or muscle volume	• Expensive • Largely confined to research and laboratory settings • Time consuming • High radiation exposure makes it unsafe for repeated measures • Cumbersome to estimate total-body muscle mass
MRI	• Excellent validity • Noninvasive • Safe • Can be used to measure CSA and/or muscle volume	• Expensive • Largely confined to research and laboratory settings • Time consuming • Cumbersome to estimate total-body muscle mass
Biopsy	• Provides insight into development at the microstructural level • Can estimate changes in CSA of different muscle fiber types • Can be used to analyze differences between protein subfractions (sarco-plasmic versus contractile)	• Invasive • Uncomfortable • Specific to a small region of muscle tissue and thus results obtained by a single biopsy do not necessarily reflect those of the entire muscle • Depth of needle penetration can influence findings

Abbreviations: MRI = magnetic resonance imaging; CT = computerized tomography; CSA = cross-sectional area; FFM = fat-free mass.

useful indirect method, both in terms of accuracy and its ability to provide segmental estimates of morphology. BIA also can be a valuable tool and has the added benefit of being able to estimate body water. However, qualitative differences between units and brands must be considered when attempting to draw relevant practical conclusions from data. The 4C method provides the greatest accuracy as a whole-body proxy of muscle mass but lacks the ability to provide regional-specific information. Given wide-ranging differences in the distribution of body fat and muscle mass between ethnicities (46), the use of proper ethnic-specific formulas are imperative for accurately determining body composition across populations.

Of the site-specific methods, circumference measurements have the least ability to estimate muscle growth when used as a standalone assessment. However, when combined with

other methods such as skinfold measurement, it can provide valid, useful data. Imaging techniques (i.e., ultrasound, CT, and MRI) are excellent assessment options, but each has drawbacks that require consideration. Muscle biopsies have several inherent limitations, but they provide unique insights into fiber type-specific responses and important hypertrophic implications that cannot be gleaned from other methods.

It is important to understand that these methods don't always show consistency in the magnitude of estimated exercise-induced hypertrophy. However, by taking into account that the various estimates represent different hypertrophic constructs, findings can be put in the proper context. A complete picture of hypertrophy can be obtained only when combining multiple types of assessments and interpreting their results as a whole.

TAKE-HOME POINTS

- No single measurement tool provides comprehensive insights into muscle hypertrophy and its associated changes over time.
- All methods of measurement offer advantages and disadvantages.
- Indirect measures of hypertrophy lack the ability to detect subtle changes in muscle mass over time.
- Muscle volume provides an estimate of whole-muscle changes but does not account for potential regional hypertrophic differences.
- Muscle biopsy is the only method capable of providing information about fiber type-specific hypertrophy and differences between protein subfractions (sarcolemmal versus contractile) within muscle.
- Combining multiple types of methods is needed to provide a complete picture of muscle development in a given individual or group of individuals.